Exercise C1

Q1. Let $\Delta_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{32} & a_{32} & a_{33} \end{vmatrix}$ and let Δ_1 denote the determinant formed by the cofactors of elements of Δ_0 and Δ_2 denote the determinant formed by the cofactor at Δ_1 similarly Δ_n denotes the determinant formed by the cofactors at Δ_{n-1} then the determinant value of Δ_n is

(A) Δ_0^{2n} (B) $\Delta_0^{2^n}$ (C) $\Delta_0^{n^2}$ (D) Δ_0^2

Ans. B

Sol. $\Delta_1 = \Delta_0^2$

$\Delta_2 = \Delta_1^2 = \Delta_0^4$

$\Delta_3 = \Delta_2^2 = \Delta_8^0$

$\Delta_4 = \Delta_3^2 = \Delta_0^{16}$

and so on $\Delta_n = \Delta_0^{2^n}$

Q2. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose det. A = 6. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then

(A) det. B = 6 (B) det. B = – 6 (C) det. B = 12 (D) det. B = – 12

Ans. C

Sol. Consider the det. B, using $R_1 \to R_1 + R_2 + R_3$

$$B = 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$$

using $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

$$= 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ -p & -q & -r \\ -x & -y & -z \end{vmatrix}$$

using $R_1 \to R_1 + R_2 + R_3$

B = 2 det. A = 2 . 6 = 12

Q3. The system of equations :

$2x\cos 2q + y\sin 2q - 2\sin q = 0$
$x\sin 2q + 2y\sin 2q = -2\cos q$
$x\sin q - y\cos q = 0$, for all values of q, can

(A) have a unique non - trivial solution (B) not have a solution

(C) have infinite solutions (D) have a trivial solution

Ans. -

Sol. Slpe of (1) and (2) is $\cot\theta \Rightarrow$ (1) and (2) are parallel and slope of (3) is $\tan\theta$

$\Rightarrow$ no solution

Using $R_2 \to R_2 - (2\cos\theta) R_3$ and $R_1 \to R_1 + (2\sin\theta)R_3$, the value of determinant is 4.

Q4. If the system of equations $x - Ky - z = 0$, $Kx - y - z = 0$ and $x + y - z = 0$ has a non zero solution, then the possible values of K are

(A) $-1, 2$ (B) $1, 2$ (C) $0, 1$ (D) $-1, 1$

Ans. D

Sol. -

Q5. The system of equations

$x - y\cos\theta + z\cos 2\theta = 0$
$-x\cos\theta + y - z\cos\theta = 0$
$x\cos 2\theta - y\cos\theta + z = 0$
has non trivial solution for θ equals

(A) $n\pi$ only, $n \in I$ (B) $n\pi + \frac{\pi}{4}$ only, $n \in I$ (C) $(2n-1)\frac{\pi}{2}$ only, $n \in I$

(D) all value of θ

Ans. D

Sol. for non trivial solution

$$\begin{vmatrix} 1 & -\cos\theta & \cos 2\theta \\ -\cos\theta & 1 & -\cos\theta \\ \cos 2\theta & -\cos\theta & 1 \end{vmatrix}$$

using $C_1 \to C_1 - C_3$

$$\begin{vmatrix} 2\sin^2\theta & -\cos\theta & \cos 2\theta \\ 0 & 1 & -\cos\theta \\ -2\cos\theta & -\cos\theta & 1 \end{vmatrix} = 0$$

$$2\sin^2\theta \begin{vmatrix} 1 & -\cos\theta & \cos 2\theta \\ 0 & 1 & -\cos\theta \\ -1 & -\cos\theta & 1 \end{vmatrix} = 0$$

$\sin^2\theta = 0$

or $1[1 - \cos^2\theta] - 1\,[\cos^2\theta - \sin^2\theta\,]$

$\sin^2\theta - \sin^2\theta = 0$

hece $D = 0\ \forall\ \theta \in R \qquad \Rightarrow \qquad (D)$

Q6. If $D(x) = \begin{vmatrix} x-1 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+1)^3 \end{vmatrix}$ then the coefficient of x in D(x) is

(A) 5 (B) –2 (C) 6 (D) 0

Ans. A

Sol. Coefficient of x in D (x) = D' (0)

Q7. The set of equations
$\lambda x - y + (\cos\theta)\, z = 0,\quad 3x + y + 2z = 0,\ (\cos\theta)x + y + 2z = 0$
$0 \le \theta < 2\pi$, has non- trivial solution(s)

(A) for no value of λ and θ (B) for all value of λ and θ

(C) for all value of λ and only two values of θ

(D) for only one value λ and all values of θ

Ans. A

Sol. $D = \cos\theta - \cos^2\theta + 6 \neq 0$ sicne $D \neq 0 \Rightarrow$ only trivial solution is possible $\Rightarrow$ (A)

Q8. The system of equations $(\sin\theta)x + 2z = 0$, $(\cos\theta)x + (\sin\theta)y = 0$, $(\cos\theta)y + 2z = a$ has

(A) no unique solution (B) a unique solution which is a function of a and θ

(C) a unique solution which is independent of a and θ

(D) a unique solution which is independent of θ only

Ans. B

Sol. $$D = \begin{vmatrix} \sin\theta & 0 & 2 \\ \cos\theta & \sin\theta & 0 \\ 0 & \cos\theta & 2 \end{vmatrix}$$

$\sin\theta\,(2\sin\theta) + 2\cos^2\theta = 2;\ D_1 = a\sin\theta$

Q9. The equation $\begin{vmatrix} (1+x)^2 & (1-x)^2 & -(2+x^2) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} + \begin{vmatrix} (1+x)^2 & 2x+1^2 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x-2 & 2x-3 \end{vmatrix} = 0$

(A) has no real solution (B) has 4 real solutions

(C) has two real and two non-real solutions

(D) has infinite number of solutions , real or non-real

Ans. D

Sol. 1st two columns of 1st determinant are same as 1st two rows of 2nd. Hence transpose the 2nd. Add the two determinants and use $C_1 \to C_1 + C_3 \Rightarrow D = 0$

Q10. Let $\Delta_1 = \begin{vmatrix} ap^2 & 2ap & 1 \\ aq^2 & 2aq & 1 \\ ar^2 & 2ar & 1 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} apq & a(p+q) & 1 \\ aqr & a(q+r) & 1 \\ arp & a(r+p) & 1 \end{vmatrix}$ then

(A) $\Delta_1 = \Delta_2$ (B) $\Delta_1 = 2\Delta_2$ (C) $\Delta_1 = 2\Delta_2$ (D) $\Delta_1 = 2\Delta_2 = 0$

Ans. D

Sol. -

Q11. The value of λ for which the system of equations $2x - y - z = 12$, $x - 2y + z = -4$, $x + y + lz = 4$ has no solution is

(A) 3 (B) –3 (C) 2 (D) –2

Ans. D

Sol. -

Q12. The value of $\begin{vmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{vmatrix}$ is independent of

(A) α (B) β (C) both α and β (D) neither α nor β

Ans. A

Sol. -

Q13. The values of θ, λ for which the following equations

$\sin\theta\, x - \cos\theta\, y + (\lambda+1)z = 0$; $\cos q x + \sin\theta y - \lambda z = 0$; $\lambda x + (\lambda + 1)y + \cos\theta\, z = 0$

have non trivial solution, is

(A) $\theta = n\pi$, $\lambda \in R - \{0\}$ (B) $\theta = 2n\pi$, λ is any rational number

(C) $\theta = (2n + 1)\pi$, $\lambda \in R^+$, $n \in I$ (D) $\theta = (2n + 1)\frac{\pi}{2}$, $\lambda \in R$, $n \in I$

Ans. D

Sol. For non trivial solution = 0; this gives $2\cos\theta(\lambda^2+\lambda+1)=0$

Q14. Number of triplets of a, b & c for which the system of equations,

$ax - by = 2a - b$ and $(c+1)x + cy = 10 - a + 3b$

has infinitely many solutions and $x = 1$, $y = 3$ is one of the solutions, is :

(A) exactly one (B) exactly two (C) exactly three (D) infinitely many

Ans. B

Sol. put $x = 1$ & $y = 3$ in 1st equation $\Rightarrow a = -2b$ & from 2nd equation

$c = \frac{9+5b}{4}$; Now use $\frac{a}{c+1} - \frac{b}{c} = \frac{2a-b}{10-a+3b}$; from first two $b = 0$ or $c = 1$; if $b = 0 \Rightarrow a = 0$ & $c = \frac{9}{4}$; if $c = 1$; $b = -1$; $a = 2$

Q15. For any ΔABC, the value of determinant $\Delta \begin{bmatrix} \sin^2 A & \cos A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{bmatrix}$ is –

(A) 0 (B) 1 (C) sin A sin B sin C (D) sin A + sin B + sin C

Ans. A

Sol. -

Q16. If a, b, c are non-zero real numbers, then $\begin{bmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{bmatrix}$ is equal to

(A) abc (B) $a^2b^2c^2$ (C) ab + bc + ca (D) None of these

Ans. D

Sol.

$$D = \begin{bmatrix} ab^2c^2 & abc & a(b+c) \\ bc^2a^2 & bca & b(c+a) \\ ca^2b^2 & cab & c(a+b) \end{bmatrix}$$

$(R_1 \to aR_1, R_2 \to bR_2, R_3 \to cR_3)$

$$= abc \begin{vmatrix} bc & 1 & ab+ac \\ ca & 1 & bc+ab \\ ab & 1 & ca+cb \end{vmatrix}$$

$$= abc\,(ab+bc+ca) \begin{vmatrix} bc & 1 & 1 \\ ca & 1 & 1 \\ ab & 1 & 1 \end{vmatrix} (C_3 \to C_3 + C_1) = 0$$

Q17. The determinant $\begin{vmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix}$ is :

(A) 0 (B) independent of θ (C) independent of ϕ (D) independent of both θ and ϕ

Ans. B

Sol. $$\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix} = \frac{1}{\sin\phi\cos\phi}\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin\theta\sin\phi & \sin\phi\cos\theta \\ -\cos\theta\cos\phi & \sin\theta\cos\phi \end{vmatrix}$$

Applying $R_1 \to R_1 + R_2 + R_3$ ($R_1 \to R_1 + R_2 + R_3$)

$$= \frac{1}{\sin\phi\cos\phi}\begin{vmatrix} 0 & 0 & 2\cos^2\phi \\ \sin\theta\sin\phi & \sin\varphi\cos\theta & \sin^2\phi \\ -\cos\theta\cos\phi & \sin\theta\cos\phi & \cos^2\phi \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 2\cos^2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix} = 2\cos^2\phi\,(\sin^2\theta + \cos^2\theta) = 2\cos^2\phi$$

Q18. The number of distinct real roots of the equation $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. $$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$

$C_1 \to C_1 - C_2$ $C_2 \to C_2 - C_3$

$$\begin{vmatrix} \sin x - \cos x & 0 & \cos x \\ -(sinx - cosx) & \sin x - \cos x & \cos x = 0 \\ 0 & -(sinx - cosx) & \sin x \end{vmatrix}$$

$$(\sin x - \cos x)^2 \begin{vmatrix} 1 & 0 & \cos x \\ -1 & 1 & \cos x \\ 0 & -1 & \sin x \end{vmatrix} = 0$$

$$(\sin x - \cos x)^2\,[1(sinx + cosx) + cosx(1)] = 0$$

$\sin x = \cos x$ $\sin x + 2\cos x = 0$

$\tan x = 1$ $\tan x = -2$

$x = \pi / 4$

Q19. If the determinant $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ is expanded in powers of sin x, then the constant term is :

(A) 0 (B) 1 (C) – 1 (D) 2

Ans. C

Sol. $\cos 2x(\cos^2 2x - \cos^4 x) - \sin^2 x(\sin^2 x \cos 2x - \cos 4x \cos^2 x) + \cos 4x(\sin^2 x \cos^2 x - \cos 4x \cos 2x)$

$\Rightarrow \cos^3 2x + 2 \cos 4x \cos^2 x \sin^2 x - \cos 2x \cos^4 x - \sin^4 x \cos 2x + \sin^2 x \cos^2 x \cos 4x - \cos^2 4x \cos 2x$

$\Rightarrow (1 - 2\sin^2 x)^3 + 2(2(1 - \sin^2 x) - 1) \sin^2 x (1 - \sin^2 x) - [(1 - 2\sin^2 x)(2(1 - 2\sin^2 x) - 1)^2 + (1 - \sin^2 x)^2 (1 - 2\sin^2 x) + \sin^4 x(1 - 2\sin^2 x)]$

$1 + 0 - (1 + 1 + 0)$

$1 - 2 = -1$

Q20. The determinant $\begin{vmatrix} a^2 & a^-(b-c)^2 & bc \\ b^2 & b^2-(c-a)^2 & ca \\ c^2 & c^2-(a-b)^2 & ab \end{vmatrix}$ is divisible by :

(A) a + b + c (B) (a + b) (b + c) (c + a) (C) $a^2 + b^2 + c^2$

(D) (a - b) (b - c) (c - a)

Ans. A, C, D

Sol. Use $C^2 \to C^2 - C^1 - 2C^3$ then $C^1 \to C^1 + C^2$ take $a^2 + b^2 + c^2$ common from first column

Q21. Which of the following determinant(s) vanish(es)?

(A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$ (B) $\begin{vmatrix} 1 & ab & \frac{1}{a}+\frac{1}{b} \\ 1 & bc & \frac{1}{b}+\frac{1}{c} \\ 1 & ca & \frac{1}{c}+\frac{1}{a} \end{vmatrix}$ (C) $\begin{vmatrix} a & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$

(D) $\begin{vmatrix} log_x xyz & log_x y & log_x z \\ log_y xyz & 1 & log_y z \\ log_z xyz & log_z y & 1 \end{vmatrix}$

Ans. A, B, C, D

Sol. -

Q22. If $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$ then $\theta =$

(A) $\frac{7\pi}{24}$ (B) $\frac{5\pi}{24}$ (C) $\frac{11\pi}{24}$ (D) $\frac{\pi}{24}$

Ans. A, C

Sol. $R_1 \to 2R_1 - R_2$ and $R_2 \to R_2 - R_3$